

VU Machine Learning

Winter semester 2025

Exercise 2

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- Groups of 3 students
- Implement two techniques for regression
- Perform experiments and compare to existing/other techniques
- Submit the source code
- Prepare a slide presentation
 - Around 20-40 slides, including tables & diagrams
 - No report is needed for this assignment (unless you prefer to write one)
- Submission: 15.12.2025, 14:00
- Discussions: 16.12. – 19.12. or 8.1. – 9.1.

- Pick 3 regression data sets
 - Data sets from UCI ML Repository, Kaggle... that were published after 2020. You can also use one data set from the previous assignment
- With different characteristics
 - number of samples – small vs. large
 - number of dimensions – low vs. high dimensional

- Implement a [regression tree algorithm](#) and a [random forest algorithm](#) (based on the implemented regression tree algorithm) for predicting numeric values
 - You can find various implementations for these algorithms. However, you can also apply your own ideas for splitting of instances
 - You should implement these algorithms from scratch
- Please do not use any part of existing code
- You can use existing code/functions for general parts like
 - Code for reading the input and testing the algorithm (cross- validation, performance metrics for regression...)

- Compare your implemented techniques with the existing implementations of regression trees/random forest and one other existing regression techniques
 - You can use the default parameters for the existing techniques
- Experiment with at least three configurations (number of trees, ...) for random forest
- Use at least two performance metrics for comparison
- Apply cross-validation
- Conclusions
 - How efficient are your algorithms
 - Performance of your algorithms
 - Other findings

A zip file with

- **Source code:**
 - You can use any programming language: Python, Matlab, R...
 - Provide the information for the packages needed to run your code
- **Data sets (links to the data sets)**
- **Slides**
 - Around 20 - 40 slides, including tables & diagrams
 - No report needed
- Submission deadline: 15.12.2025, 14:00

- Characteristics of data sets
- Details regarding the implementation (pseudocode...)
 - Main issues regarding the implementation (lessons learned)
 - Experiments and performance metrics used
- Comparison to other techniques
- Discussion of experimental results and comparison across the different datasets and techniques (tables, figures)
- Conclusions/lessons learned

- Length of discussions
 - 15 minutes
- You can use the slides you submitted and skip some of them during the discussion
- You will also be asked questions about your source code, implemented algorithms and performance metrics
- Every group member should be prepared to answer questions about all parts of the assignment

The evaluation is based on the submission and answers to questions during the individual discussion:

- Implementation of algorithms and experiments: 65%
- The choice of data sets and comparison to other techniques: 15%
- Conclusions, lessons learned: 20 %